

Python Interview Questions with Answers

1. What is Python?

Python is a high-level, interpreted, and object-oriented programming language known for its simple syntax and readability.

2. What are the features of Python?

Easy to learn, simple syntax, interpreted language, object-oriented, cross-platform, large libraries support, supports AI and web development.

3. Difference between interpreted and compiled language?

Interpreted languages execute line by line while compiled languages convert whole code first.

4. What are Python data types?

int, float, string, list, tuple, dictionary, set, boolean.

5. Difference between List and Tuple?

List is mutable and uses [], while Tuple is immutable and uses ().

6. What is a dictionary in Python?

Dictionary stores data in key-value pairs.

7. Difference between == and = ?

= is assignment operator while == is comparison operator.

8. What is a function in Python?

A function is a reusable block of code.

9. What is OOP in Python?

OOP stands for Object-Oriented Programming with concepts like class, object, inheritance, polymorphism, and encapsulation.

10. What is a class and object?

Class is a blueprint while object is an instance of class.

11. What is inheritance?

Inheritance allows one class to use properties of another class.

12. What is encapsulation?

Encapsulation means hiding internal data and controlling access.

13. What is polymorphism?

Polymorphism means one function behaves differently in different situations.

14. Difference between append() and extend()?

append() adds a single element while extend() adds multiple elements.

15. What is exception handling?

Exception handling manages runtime errors using try, except, and finally.

16. Difference between remove(), pop(), and del?

remove() removes a value, pop() removes by index, and del deletes an object.

17. What is lambda function?

Lambda is an anonymous one-line function.

18. What are modules in Python?

Modules are Python files containing functions and variables.

19. Difference between break and continue?

break stops the loop while continue skips current iteration.

20. What is list comprehension?

A short way to create lists.

21. Difference between deep copy and shallow copy?

Shallow copy copies references while deep copy copies complete objects.

22. What is `__init__` in Python?

`__init__` is a constructor method automatically called when an object is created.

23. What is recursion?

Recursion is when a function calls itself.

24. What is PEP 8?

PEP 8 is Python coding style guideline for writing clean and readable code.

25. Where is Python used?

Python is used in AI, Machine Learning, Web Development, Automation, Data Science, Cybersecurity, and Cloud Computing.